

SURNAME

NAME

MAT. NUMBER

Signature _____

1

Consider the following DAE system.

$$\begin{cases} x'(t) - z'(t) = t^2 \\ x(t)y(t) = 1 \\ x(t)y(t)z(t) = t \end{cases}$$

rewrite in the form

$$\boldsymbol{E}(\boldsymbol{x}, t)\boldsymbol{x}' = \boldsymbol{G}(\boldsymbol{x}, t)$$

perform index reduction

$$\boldsymbol{E}^{(1)}(\boldsymbol{x}, t)\boldsymbol{x}' = \boldsymbol{G}^{(1)}(\boldsymbol{x}, t), \qquad \dots \qquad \boldsymbol{E}^{(k)}(\boldsymbol{x}, t)\boldsymbol{x}' = \boldsymbol{G}^{(k)}(\boldsymbol{x}, t),$$

until $\boldsymbol{E}^{(k)}(\boldsymbol{x}, t)$ is non singular and DAE is reduced to an ODE. Compute:

Differential index:

Invariants at each index reduction step:

Matrices $\boldsymbol{E}^{(j)}(\boldsymbol{x}, t)$ and vector $\boldsymbol{G}^{(j)}(\boldsymbol{x}, t)$ at each step

2

Consider the following DAE system:

$$\begin{cases} x''(t) = \lambda(t)y(t) + \sin(t) \\ y''(t) = \lambda(t)x(t) + \cos(t) \\ y(t) = x(t)^2 - 1 \end{cases}$$

reduce to order 1 then using dummy variable reduce the DAE to index 1. Compute:

Differential index:

DAE rewritten using dummy variables

Invariants at each index reduction step:

DAE reduced to index 1

Projection to invariants problem: